Institution's Innovation Council
MHRD's Innovation Cell, AICTE

Innovation / Prototype
(Product/Service/Process)

Team Details

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Add more fields if required

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Add more fields if required

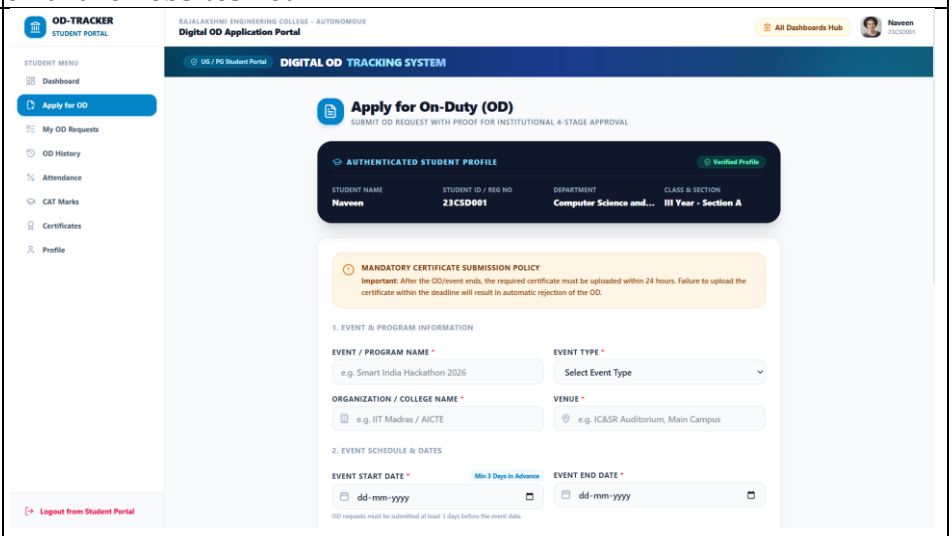
Institute Name and Address:
Rajalakshmi Engineering College, Thandalam, Chennai, Tamil Nadu.

Title (20 Words Max)	OD Tracking Application – A Digital On-Duty Request, Approval and Certificate Tracking System
***Developed as part of**	Design Thinking and Innovation Project
***Choose the Financial Year, during the Idea-PoC/Innovation Developed**	2026–2027
***Sector / Domain**	Information and Communication Technology (ICT)
Innovation Type	Product Innovation

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***Development Stage - Technology Maturity of the Solution/Innovation in terms of Technology Readiness Level TRL (if applicable (**Refer TRL Stages**)**	TRL 6 – Prototype system tested in the intended environment, demonstrating functionality close to its expected performance.
Development Stage - Manufacturing Maturity of the Solution/Innovation in terms of Manufacturing Readiness Level (MRL) Refer MRL Stages	Not Applicable
Development Stage: Investment Readiness Level of the Solution/Innovation (IRL). Refer IRL Stages	IRL 4 – Prototype Low-Fidelity Minimum Viable Product (MVP).
***Define the problem and its relevance to today's market / society / industry need (Max: 100 Words)**	The existing On-Duty (OD) approval process in educational institutions is largely manual and paper-based. Students must submit physical OD letters, resulting in delays, document misplacement and difficulties in tracking application status. Faculty members and HODs face challenges in managing multiple requests, verifying certificates and maintaining records. The absence of a centralized system increases administrative workload and communication gaps. A digital solution is required to simplify OD management, improve transparency, reduce paperwork and ensure efficient tracking of student requests and certificate verification.
***Describe the Solution / Proposed / Developed (Max: 100 Words)**	The OD Tracking Application is a web-based platform that digitizes the complete On-Duty management process. Students can submit OD requests with supporting documents, track approval status and upload participation certificates. Faculty members and HODs can review requests, approve or reject applications, add remarks and verify certificates. Email notifications keep students informed about application updates. The system maintains centralized records of OD requests, approvals and certificates, improving communication, transparency and administrative efficiency within educational institutions.
***Explain the uniqueness and distinctive features of the (product / process / service) solution (Max: 100 Words)**	The application provides a centralized digital platform for managing the complete OD lifecycle, from request submission to final certificate verification. Its role-based interfaces support students, faculty members and HODs with different responsibilities. Key features include digital document submission, approval workflows, email notifications, certificate uploading, verification tracking and centralized record management. It reduces manual intervention, minimizes paperwork and provides transparent status tracking, making OD management more organized and efficient.

***How your proposed / developed (product / process / service) solution is different from similar kind of product by the competitors if any (Max: 100 Words)**	Unlike traditional paper-based OD management systems, this application integrates request submission, faculty approval, HOD review, email notifications and certificate verification into a single platform. It provides role-based access for students and administrators, centralized document storage and searchable OD records. The system is designed specifically for the requirements of educational institutions, simplifying coordination between students, faculty members and department heads while reducing administrative effort.
Is there any IP or Patentable Component associated with the Solution?	No patent filed currently.
Has the Solution Received any Innovation Grant/Seed fund Support?	No.
Are there any Recognitions (National/International) Obtained by the Solution?	No.
Is the Solution Commercialized either through Technology Transfer or Enterprise Development/Startup?	No.
Had the Solution Received any Pre-Incubation/Incubation Support?	No.
***Utility: Highlight the utility/value proposition (key benefits) aspects of the solution/innovation* (Max: 100 Words)**	The application reduces paperwork and administrative workload by digitizing OD requests and approvals. Students benefit from convenient online submission, real-time status tracking and simplified certificate uploads. Faculty members and HODs can efficiently manage approvals, verify certificates and maintain organized records. Email notifications improve communication and reduce unnecessary follow-ups. The platform provides transparency, accountability and centralized access to OD-related information, making the overall process faster and more reliable.

***Scalability: Highlight the market potential aspects of the Solution/Innovation (Potential Market Size, segmentation and Target users/customers etc.) (Max: 100 Words)**	The OD Tracking Application can be implemented across multiple departments and educational institutions. Its role-based architecture allows adaptation to different institutional approval structures. The platform can be enhanced with mobile application support, automated reminders, QR-based event verification, attendance integration and advanced reporting features. Educational institutions, engineering colleges, universities and other academic organizations represent potential target users. Its digital architecture supports expansion to accommodate increasing numbers of students and OD requests.
***Economic Sustainability: Highlight commercialisation/business application aspects of the solution (how it is going to economic profitable and viable) (Max: 100 Words)**	The application can be economically sustainable through a software-as-a-service (SaaS) model or institutional licensing. Educational institutions can adopt the platform to reduce paperwork, administrative effort and document management costs. A centralized digital system minimizes repetitive manual work and improves operational efficiency. Future revenue opportunities may include annual institutional subscriptions, customization services, maintenance and technical support. The use of scalable software infrastructure can help control operational expenses.
***Environmental Sustainability: Highlight environmental friendliness aspects and related benefit of the solution/innovation (Max: 100 Words)**	The application promotes environmental sustainability by replacing physical OD request forms, printed approval letters and paper-based certificate records with digital documents. Students and faculty members can submit, review and verify documents online, reducing paper consumption and printing requirements. Centralized digital storage also minimizes physical record maintenance. By encouraging paperless administrative operations, the application contributes to resource conservation and environmentally responsible institutional management.
Video URL	rajalakshmi-odtracker-b3b2g6d7aahxcagv.southeastasia-01.azurewebsites.net
Upload Photograph: (JPG, PNG, PDF max 2 MB)	

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(e) Timeline- Develop a timeline against the milestones for taking Idea/PoC to Prototype Development and (or) Commercial level/start-up stage.	**Phase** — **Milestone** Phase 1 — Problem identification and requirement analysis Phase 2 — UI/UX design and database planning Phase 3 — Frontend and backend development Phase 4 — OD approval workflow and email integration Phase 5 — Certificate upload and verification Phase 6 — Testing, deployment and improvements
Applicability of Solution 10 Marks *(Max 50 words for each from a-e)*	
(a) **Usability:** what is the usability of your innovation. *(Level of acceptance of innovation and its Features among target group)*	The application provides an easy-to-use interface for students, faculty members and HODs. Students can submit OD requests and track their status, while faculty members can manage approvals and certificate verification through dedicated dashboards.
(b) **Scalability:** how your innovation will be scalable at market level.	The application can be extended across multiple departments and educational institutions. Additional features such as mobile applications, QR-based verification and analytics can support increasing users and institutional requirements.
(c) **Economic sustainability:** Explain the potential of innovation to become profitable or financially viable.	The solution can reduce administrative costs, paperwork and manual effort. Future financial viability can be achieved through institutional subscriptions, licensing, customization and maintenance services.
(d) **Environment Sustainability:** How your innovation is environment friendly or address environmental problems.	The platform reduces paper usage by digitizing OD applications, approval letters and certificates. Online document management minimizes printing and physical record storage.
(e) Is there any Intellectual Property (IP) Component associated with innovation? if yes, explain.	Currently, no patent has been filed for the application. The software code, user interface and workflow implementation represent potential intellectual property components that may be considered for future protection.
Define the potential market size (in terms of INR) and target customers. *(Max 100 words)*	The potential market includes engineering colleges, universities, autonomous institutions and other educational organizations that manage student On-Duty requests. Target customers include institutional management, academic departments and administrative offices. The solution can be offered through institutional licensing or subscription-based services. The market potential in INR requires further validation through research on the number of target institutions, expected adoption rate and pricing. The application can initially target colleges in Tamil Nadu before expanding to other regions.

Annexure 1

Themes:

1. Healthcare & Biomedical devices.
2. Agriculture & Rural Development.
3. Smart Vehicles/ Electric vehicle/ Electric vehicle motor and battery technology.
4. Food Processing.
5. Robotics and Drones.
6. Waste management.
7. Clean & Potable water.
8. Renewable and affordable Energy.
9. IoT based technologies (e.g. Security & Surveillance systems etc)
10. ICT, cyber physical systems, Block chain, Cognitive computing, Cloud computing, AI & ML.

9 stages of TRL:

TRL 0 : Idea. Unproven concept, no testing has been performed.

TRL 1 : Basic research. Principles postulated observed but no experimental proof available. TRL 2 : Technology formulation. Concept and application have been formulated.

TRL 3 : Applied research. First laboratory tests completed; proof of concept.

TRL 4 : Small scale prototype built in a laboratory environment ("ugly" prototype). TRL 5 : Large scale prototype tested in intended environment.

TRL 6 : Prototype system tested in intended environment close to expected performance. TRL 7 : Demonstration system operating in operational environment at pre-commercial scale. TRL 8 : First of a kind commercial system. Manufacturing issues solved.

TRL 9 : Full commercial application, technology available for consumers.